

Nuclear Medicine

Foundations, Activity and Radioactive Decay

Study Notes

How to use this edition

Read Sections 1–7 for the foundations. Sections 8–10 extend the material into clearance, production and image formation. The final sections provide corrected terminology, worked practice and a compact formula sheet.

The explanations emphasize precise terminology, clearly stated assumptions and consistent units. Worked examples connect the physical principles to practical calculations, while references support further study.

Using the calculations. Follow each derivation before substituting numerical values. Rounded teaching data are distinguished from evaluated nuclear data, and supplementary sections provide context for further study.

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1 Recording map and the purpose of nuclear medicine physics

1.1 Source coverage

All four relevant recordings found in the available Plaud library were used. The library listing contained 14 recordings; the remaining titles concerned other courses. All 469 transcript segments in the four selected recordings were retrieved, including every continuation page.

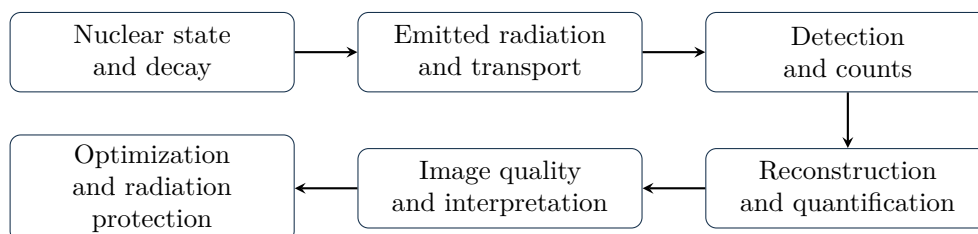
ID / date	Recording length	Topics used
L1 / 7 Sep	86 min 32 s	Nuclear Medicine Physics Foundations and Course Orientation: course scope; physicist's role; atomic structure; quantum numbers; quarks; fundamental forces.
L2 / 10 Sep	94 min 31 s	Fundamental Forces, Standard Model, and the Atomic Nucleus: particles; atomic models; nuclide notation; mass units; stability chart; nuclear binding energy and exercises.
L3 / 14 Sep	46 min 19 s	Radioactive Decay and Half-Life: differential equation; exponential law; half-life derivation; statistical decay; parent and daughter nuclei.
L4 / 17 Sep	65 min 17 s	Radioactive Decay Kinetics, Nuclear Medicine Dosing, and Course Logistics: mean life; specific activity; weight-based activity; decay modes; fission; conservation laws.

L1 has 94 segments, L2 107, L3 121 and L4 147. Recording identifiers and coverage notes are retained in Appendix A. Durations are recording metadata; timestamps refer to elapsed recording time.

1.2 The physical chain

Recording basis: L1, approximately 11:00–31:00 and 62:00–73:00.

Nuclear medicine uses radionuclides to investigate physiology and disease, and in selected applications to deliver therapy. A radiopharmaceutical combines a radionuclide with a chemical form that determines how it distributes in the body. Understanding the radionuclide alone is insufficient: chemical targeting, uptake and clearance determine where its radiation originates.



The medical physicist connects these stages: measuring activity, characterizing instruments, assessing quantitative accuracy, investigating artifacts and optimizing examinations. Physicians provide clinical interpretation and medical decisions; technologists perform procedures and assigned quality-control tasks; engineers service equipment. Responsibilities overlap and depend on local training and governance.

1.3 Course roadmap: what was previewed

The orientation named six broad areas: nuclear foundations; radiation interactions; planar and gamma camera imaging; SPECT; PET and hybrid imaging; quantitative imaging, QA/QC, protection and theranostics. Detailed reconstruction, SUV, partial-volume effects and instrument testing were *announced topics*, not completed lectures in these recordings. Section 9 offers a concise supplementary introduction.

A source continues to decay when an instrument is switched off. This makes time, distance, shielding and contamination control central to unsealed-source work. Optimization seeks image quality sufficient for the clinical task with appropriate activity and acquisition settings. Activity, absorbed dose and image quality are connected but are different quantities.

2 Atoms, nuclei and notation

Recording basis: L1, 73:00–78:00; L2, 37:00–71:00.

2.1 Atomic models and electronic structure

Dalton's atomic model, Thomson's electron-containing model, Rutherford's compact nucleus and Bohr's quantized orbits form a historical sequence. Modern quantum mechanics describes electrons using orbitals and probability distributions. Bohr's model is useful for hydrogen-like systems, but literal circular orbits do not describe general many-electron atoms.

The nucleus contains protons and neutrons, jointly called nucleons. It contains almost all atomic mass; electrons occupy the much larger surrounding region. A neutral atom has as many electrons as protons. Removing an electron creates a positive ion; adding one can create a negative ion. Ionization changes electronic charge without necessarily changing nuclear composition.

Particle	Charge	Approximate mass (u)	Rest energy (MeV)
Electron	$-e$	0.00054858	0.510999
Proton	$+e$	1.00727647	938.272
Neutron	0	1.00866492	939.565

Values are rounded for study. A hydrogen-1 *atom* has mass about 1.00782503 u; this differs from the proton mass because it includes an electron, less electronic binding energy.

2.2 The four electron quantum numbers

Symbol	Allowed values	Meaning
n	$1, 2, 3, \dots$	Principal shell; K, L, M correspond to $n = 1, 2, 3$.
ℓ	$0, 1, \dots, n - 1$	Orbital angular momentum; s, p, d, f correspond to $\ell = 0, 1, 2, 3$.
m_ℓ	$-\ell, \dots, 0, \dots, +\ell$	Orbital angular-momentum projection; $2\ell + 1$ values.
m_s	$-\frac{1}{2}, +\frac{1}{2}$	Spin projection for a spin- $\frac{1}{2}$ electron.

Precision note. Three orbital orientations apply to the p subshell, not every subshell. Electron spin is intrinsic angular momentum, not a tiny sphere rotating mechanically. Valence electrons are associated with outer-shell chemistry; K-shell electrons are inner-shell electrons.

The Pauli principle permits at most two electrons with opposite spin projections in a given spatial orbital. In many-electron atoms, energy depends on more than n alone because of screening and electron interactions.

2.3 Nuclide notation and nuclear families

Write a nuclide as ${}^A_Z\text{X}$, where

$$A = Z + N_n, \quad Z = \text{proton number}, \quad N_n = \text{neutron number}.$$

Here N_n denotes neutrons *within one nucleus*; later, $N(t)$ counts undecayed nuclei *in a sample*. Likewise, A in a nuclide label is mass number; $A(t)$ in a decay equation is activity. Mass number counts nucleons and is dimensionless; it is not the measured atomic mass in u.

Family	Same quantity	Example / distinction
Isotopes	Z	${}^1\text{H}$, ${}^2\text{H}$, ${}^3\text{H}$; different N_n .
Isotones	N_n	${}^2\text{H}$ and ${}^3\text{He}$ each have one neutron.
Isobars	A	${}^3\text{H}$ and ${}^3\text{He}$ each have three nucleons.
Nuclear isomers	A, Z	${}^{99m}\text{Tc}$ and ${}^{99}\text{Tc}$; different nuclear energy states.

Hydrogen-1, hydrogen-2 and hydrogen-3 are protium, deuterium and tritium. Their nuclei are the proton, deuteron and triton. Hydronium (H_3O^+) is a molecular ion, not another name for hydrogen-1.

An asterisk indicates an excited nuclear state. The suffix m indicates a metastable nuclear state, which persists long enough to be identified separately. An ionic charge is placed at upper right, for example ${}^{23}_{11}\text{Na}^+$. Isotopes have closely similar electronic chemistry, although isotope effects occur; their nuclear decay properties can be very different.

2.4 Size and radiation terminology

A useful nuclear-radius estimate is

$$R \simeq r_0 A^{1/3}, \quad r_0 \simeq 1.2\text{--}1.3 \text{ fm}, \quad 1 \text{ fm} = 10^{-15} \text{ m}.$$

The lecture used 1.3 fm. The exponent acts on *mass number*, not atomic number. This is an approximate nuclear-size model, particularly crude for very light nuclei.

Ionizing radiation can remove bound electrons through its interactions. Alpha and beta particles are directly ionizing; photons and neutrons transfer energy to secondary charged particles. Excitation raises an atom or nucleus to a higher state without necessarily ejecting a particle. Electronic excitation and nuclear excitation involve different structures and typically different energy scales.

3 Fundamental interactions and nuclear stability

Recording basis: L1, 78:00–85:00; L2, 7:00–37:00 and 71:00–85:00; L4, 58:00–63:00.

3.1 Particles and forces: corrected framework

Electrons are elementary leptons. Protons and neutrons are composite baryons with valence-quark contents uud and udd . Up and down quark charges are $+2e/3$ and $-e/3$, respectively:

$$Q_p = 2(2e/3) - e/3 = e, \quad Q_n = 2e/3 - 2e/3 = 0.$$

These fractions describe *charge*, not spin. Quarks have spin $1/2$. Baryons have half-integer spin, while mesons have integer spin; a baryon spin of $2/3$ is not the correct classification.

Interaction	Carrier / description	Relevance here
Strong	Gluons at quark level	Binds quarks; its residual nuclear interaction binds nucleons over nuclear distances.
Electromagnetic	Photon	Atomic binding, charged-particle interactions, nuclear photon transitions and proton repulsion.
Weak	$W^\pm, Z^0$	Beta transformations and electron capture.
Gravitational	General relativity; graviton hypothetical	Negligible for ordinary nuclear calculations; not included in the Standard Model.

Precision note. Strong and weak do not mean “stable-nucleus force” and “unstable-nucleus force.” Both stable and radioactive nuclei have strong interactions. The type of transition determines the interaction involved. The four forces do not have equal intrinsic strengths. Dominance also depends on distance, energy and the system: gravity dominates many astronomical systems, while electromagnetic forces underlie chemistry and everyday contact forces.

The Higgs boson is a scalar particle associated with the Higgs field, not the common gauge carrier of all interactions. Its 2012 discovery supported the Higgs mechanism; it was not the first validation of general relativity. The Standard Model contains strong and electroweak physics without constituting a demonstrated grand-unified theory. Supersymmetry, grand unification and string-theory proposals are distinct ideas; they should not be treated as synonymous established results. See the CERN references in Section 14.

3.2 Stability is an energy and structure question

The competition between nuclear attraction, proton Coulomb repulsion, neutron–proton balance, shell structure and pairing affects stability. For many light stable nuclides $N_n \approx Z$; heavier stable nuclides generally have more neutrons than protons. Neutrons contribute nuclear binding without additional proton–proton repulsion. This trend does not mean arbitrarily adding neutrons always stabilizes a nucleus.

Pairing favors even numbers of protons and neutrons. The familiar light stable odd–odd nuclides are ${}^2\text{H}$, ${}^6\text{Li}$, ${}^{10}\text{B}$ and ${}^{14}\text{N}$. Avoid memorizing a universal count of stable nuclides from one chart: classifications depend on what decay has been observed, long-lived exceptions and whether isomers are counted.

A long half-life does not make a radioactive nuclide strictly stable. A daughter can also remain radioactive. A decay is energetically possible when total final rest energy is lower, but barriers and selection rules strongly affect the rate. Increasing binding energy per nucleon is a useful trend, not a universal test for every decay channel; use the full mass balance.

3.3 Reading a nuclide chart without reversing directions

First read the axes. For a chart with horizontal N_n and vertical Z :

Process	ΔN_n	ΔZ	Direction
α	-2	-2	Left and down
β^-	-1	+1	Left and up
β^+ or EC	+1	-1	Right and down
Gamma / internal conversion	0	0	Same nuclide position
Neutron emission	-1	0	Left
Proton emission	0	-1	Down

If the axes are interchanged, the visual directions change while the numerical changes remain the same. Simultaneous increases in N_n and Z do not describe one of the listed ordinary spontaneous decays; nuclear capture or fusion reactions must be distinguished from decay. The IAEA LiveChart is useful for checking a particular nuclide's actual branches.

3.4 Coulomb barriers and neutron interactions

A positive projectile encounters repulsion approaching a positive nucleus; quantum tunneling can permit penetration below a classical barrier. Neutrons have no electrostatic Coulomb barrier, but their interactions still depend on energy and nuclear structure. Nuclear forces do not switch off at a perfectly sharp universal surface, and electromagnetic interactions do not disappear inside the nucleus.

Neutron hazard cannot be inferred from mass alone. Neutron energy, fluence and interaction pathways matter, including recoil protons and capture radiation. Ordinary alpha emission is not automatically accompanied by neutron emission. A source such as an alpha emitter mixed with a suitable target can generate neutrons through a separate (α, n) reaction.

4 Mass, energy and the binding-energy exercises

Recording basis: L2, approximately 50:00–63:00 and 85:00–93:00.

4.1 Atomic mass unit and energy conversion

One unified atomic mass unit is one twelfth of the mass of a free, neutral carbon-12 atom at rest in its ground state:

$$1 \text{ u} \simeq 1.6605391 \times 10^{-27} \text{ kg}, \quad 1 \text{ u } c^2 \simeq 931.494 \text{ MeV}.$$

Also $c = 299\,792\,458 \text{ m s}^{-1}$ exactly and $1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}$ exactly. Therefore

$$\frac{(1.6605391 \times 10^{-27})(2.99792458 \times 10^8)^2}{1.602176634 \times 10^{-13}} \simeq 931.494 \text{ MeV}.$$

The denominator converts joules to MeV. Equivalently, $1 \text{ u} \simeq 931.494 \text{ MeV}/c^2$. Do not equate mass with energy without the c^2 conversion. The speed of light is of order 10^8 , not 10^{-8} , in metres per second.

The complete free-particle energy relation is $E^2 = p^2c^2 + m^2c^4$. At rest, $p = 0$ and $E = mc^2$. Carbon-12 is a reference convention; it is not selected because carbon exists only as isolated atoms. Chemical bonding and nuclear stability are separate concepts.

4.2 Mass defect and binding energy

For a nucleus containing Z protons and N_n neutrons,

$$\Delta m = Zm_p + N_nm_n - M_{\text{nucleus}}, \quad B = \Delta mc^2.$$

B is the energy required to separate the nucleus into free nucleons, or released when the bound system forms. Mass defect has mass units; binding energy has energy units. They are related, not numerically interchangeable quantities.

If using tabulated neutral *atomic* masses, the common teaching approximation is

$$B \simeq [Zm(^1\text{H}) + N_nm_n - M_{\text{atom}}]c^2.$$

Hydrogen atomic masses account for the Z electrons. Do not combine a bare proton mass with a neutral atomic mass without correcting the electrons. More precisely, if B_e denotes total positive electronic binding energy,

$$B_{\text{nuclear}} = [Zm(^1\text{H}) + N_nm_n - M_{\text{atom}}]c^2 + ZB_e(\text{H}) - B_e(X).$$

The electron correction is small for light nuclei but matters in precise heavy-nucleus calculations. The exercises below explicitly use the usual approximation.

4.3 Worked helium-4 calculation

Supplementary explanation / worked extension.

Using $m(^1\text{H}) = 1.00782503223 \text{ u}$, $m_n \simeq 1.008664916 \text{ u}$, and $M(^4\text{He}) = 4.00260325413 \text{ u}$,

$$\begin{aligned} \Delta m &= 2(1.00782503223) + 2(1.008664916) - 4.00260325413 \\ &\simeq 0.0303766423 \text{ u}, \\ B &\simeq 0.0303766423 \times 931.494 \simeq 28.2957 \text{ MeV}, \\ B/A &\simeq 7.0739 \text{ MeV per nucleon}. \end{aligned}$$

Precision note. The L2 transcript states an exercise target of 27.27 MeV. Standard masses give about **28.30 MeV** for helium-4. The 27.27 value may be a spoken or transcription error; it should not be forced by altering the calculation. The original slide was not available for comparison.

4.4 Solutions to the other assigned nuclides

Supplementary explanation / worked extension.

The same atomic-mass approximation gives the following independently calculated results. Atomic masses come from the NIST isotopic-mass table, not natural-element average atomic weights.

Nuclide	Z	N_n	Atomic mass (u)	B (MeV)	B/A (MeV)
^4He	2	2	4.00260325413	28.296	7.074
^7Li	3	4	7.0160034366	39.245	5.606
^{16}O	8	8	15.99491461957	127.619	7.976
^{235}U	92	143	235.0439301	1783.863	7.591

For example, lithium-7 gives $\Delta m = 3m(^1\text{H}) + 4m_n - M(^7\text{Li}) \simeq 0.0421313$ u. Uranium's entry is the uncorrected atomic-mass approximation, not a claim to millielectronvolt or even kiloelectronvolt precision in nuclear binding energy. Retain guard digits in the arithmetic and round the final answer to the accuracy of the assumptions.

4.5 Binding energy per nucleon and reaction energy

B/A helps compare average binding, and broadly rises toward the iron/nickel region before decreasing for heavier nuclei. Fusion of suitable light nuclei and fission of suitable heavy nuclei can release energy. Total B , B/A , and a particular reaction's Q are different quantities.

$$Q = \left(\sum m_{\text{initial}} - \sum m_{\text{final}} \right) c^2.$$

For an isolated spontaneous decay, positive available Q is required. Rest mass itself need not be conserved: total energy, including rest energy and kinetic energy, is conserved. Mass-number balance checks nucleons, not rest mass in kilograms.

5 Activity and the exponential decay law

Recording basis: L3, approximately 1:00–36:00 and 41:00–45:00; L4, 3:00–6:00.

5.1 Random nuclei, predictable populations

Radioactive decay is stochastic: one cannot predict the exact decay time of a particular nucleus. For a population of a single nuclear state with constant decay constant λ , the probability of decay in a small interval dt , conditional on survival, is approximately λdt .

For the expected number $N(t)$ of surviving nuclei, with no production or removal other than decay,

$$\frac{dN}{dt} = -\lambda N, \quad \boxed{A(t) = \lambda N(t) = -\frac{dN}{dt}}.$$

Activity is nonnegative. The minus sign compensates for the decrease in parent population. If a compartment also gains or loses nuclei biologically, $-dN/dt$ is *not* purely nuclear activity; the nuclear-decay rate remains $\lambda_{\text{phys}}N$.

5.2 Derivation with an explicit reference time

Separate variables and integrate:

$$\int_{N_0}^{N(t)} \frac{dN'}{N'} = -\lambda \int_{t_0}^t dt', \quad \ln \frac{N(t)}{N_0} = -\lambda(t - t_0).$$

Exponentiate both sides:

$$\boxed{N(t) = N_0 e^{-\lambda(t-t_0)}}, \quad \boxed{A(t) = A_0 e^{-\lambda(t-t_0)}}.$$

Precision note. One *applies* the exponential function to undo a logarithm; one does not multiply by e . Likewise, taking $\ln$ is not multiplication by $\ln$. N_0 is a number of nuclei and $A_0 = \lambda N_0$ is activity.

The exponent must be dimensionless. If λ is in h^{-1} , time must be in hours. For $A = \lambda N$ expressed in Bq, use λ in s^{-1} .

5.3 Physical half-life

At $t - t_0 = T_{1/2}$, $A/A_0 = 1/2$. Thus

$$\ln(1/2) = -\lambda T_{1/2}, \quad \boxed{\lambda = \frac{\ln 2}{T_{1/2}}}, \quad \boxed{T_{1/2} = \frac{\ln 2}{\lambda}}.$$

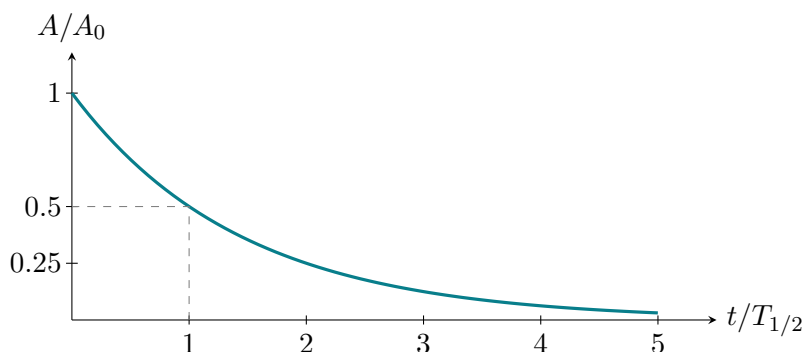
Consequently,

$$\boxed{\frac{A(t)}{A_0} = \frac{N(t)}{N_0} = 2^{-(t-t_0)/T_{1/2}}}.$$

A shorter half-life means a larger decay constant, not necessarily a larger activity unless the number of radioactive nuclei is also specified.

Elapsed half-lives	0	1	2	3	4
Fraction remaining	1	1/2	1/4	1/8	1/16
Percentage	100	50	25	12.5	6.25

After ten half-lives, $2^{-10} \simeq 0.0009766$, or 0.09766%, remains in the expectation model. This is not automatically a regulatory release criterion.



The continuous expected curve approaches zero asymptotically. A finite collection of discrete nuclei can eventually have no parent nuclei left; this does not contradict the exponential expectation. A detector can register zero counts during an interval even when the source is still radioactive. Background and detection limits further separate measurement from true activity.

5.4 Units and useful conversions

$$1 \text{ Bq} = 1 \text{ nuclear transformation per second} = 1 \text{ s}^{-1},$$

$$1 \text{ Bq} = 60 \text{ dpm}, \quad 1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq},$$

$$1 \text{ mCi} = 37 \text{ MBq}, \quad 1 \mu\text{Ci} = 37 \text{ kBq}.$$

Becquerel and hertz share dimensions of inverse time but describe different phenomena. Counts per second (cps) and disintegrations per second (dps) are not generally equal. The curie historically relates to radium-226 activity; the modern conversion above is exact.

5.5 Worked activity and scheduling calculations

Supplementary explanation / worked extension.

- 1. Activity from nuclei.** For a teaching Tc-99m half-life of 6.00 h and $N = 10^{15}$,

$$T_{1/2} = 21600 \text{ s}, \quad \lambda = 3.2090 \times 10^{-5} \text{ s}^{-1}, \quad A = 3.2090 \times 10^{10} \text{ Bq} = 32.09 \text{ GBq}.$$

This large sample activity is an arithmetic example, not a suggested patient activity.

- 2. Forward decay.** A separated sample is 800 MBq at 08:00, with 6.00 h half-life. At 14:00 it is 400 MBq; at 02:00 the following day, 18 h have elapsed and it is $800/2^3 = 100 \text{ MBq}$.

- 3. Back-correction.** To have 300 MBq after 3 h for the same half-life, the earlier activity must be

$$A_0 = A(t)e^{\lambda\Delta t} = 300 2^{3/6} = 424.26 \text{ MBq}.$$

Earlier activity is larger. Neglecting handling losses is an explicit assumption.

- 4. Solve for time.** The time to fall from A_0 to a lower A is

$$\Delta t = \frac{\ln(A_0/A)}{\lambda} = T_{1/2} \frac{\ln(A_0/A)}{\ln 2}.$$

Falling from 400 to 50 MBq with a 6 h half-life takes $6 \log_2 8 = 18 \text{ h}$.

- 5. Infer half-life.** If a decay-only source falls from 200 to 125 MBq in 4 h,

$$\lambda = \frac{\ln(200/125)}{4} = 0.11750 \text{ h}^{-1}, \quad T_{1/2} = 5.899 \text{ h}.$$

Measured count rates can substitute only when background is removed and geometry, efficiency and other response factors are unchanged.

5.6 Limits of the constant-decay assumption

Each nuclide and nuclear state has its own decay properties. Ordinary temperature and chemical changes generally do not materially alter the half-lives used in routine nuclear medicine calculations. Electron-dependent modes, especially electron capture, can be affected by electron occupancy; highly ionized systems require special treatment. A mixture does not usually follow one exponential, and daughter activity can increase through parent feeding.

6 Mean life, specific activity and administered activity

Recording basis: L4, approximately 6:00–13:00.

6.1 Mean life is not an average of isotope half-lives

For a single exponential lifetime distribution, measured from the reference time,

$$S(t) = e^{-\lambda t}, \quad p(t) = \lambda e^{-\lambda t}, \quad t \geq 0,$$

where S is survival probability and p is the normalized decay-time density. The λ factor is necessary: $\int_0^\infty p(t)dt = 1$.

$$\tau = \int_0^\infty t \lambda e^{-\lambda t} dt = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} \simeq 1.4427 T_{1/2}.$$

At $t = \tau$, the expected surviving fraction is $1/e \simeq 36.79\%$. At $T_{1/2}$ it is 50%. For the 6.00 h teaching half-life, $\tau = 8.656$ h. Mean life is useful for any exponential radionuclide, not only long-lived ones.

Precision note. The transcript’s “mean half-life” discussion blends different ideas. Mean life τ is defined for a lifetime distribution. It is not a universal effective half-life for a mixed-isotope vial.

For independent isotopes with initial activities A_{i0} ,

$$A_{\text{total}}(t) = \sum_i A_{i0} e^{-\lambda_i t}.$$

The apparent logarithmic slope changes as their proportions change. If asking for the mean remaining lifetime of a randomly selected radioactive nucleus, initial *number fractions* f_i give $\sum_i f_i / \lambda_i$; this still does not make the mixture single-exponential. Include daughter production separately where relevant.

6.2 Three denominators that must not be confused

Quantity	Expression / units	What the denominator means
Specific activity	$a_s = A/m$; Bq/g	Mass of specified material, radionuclide or chemical preparation; state the convention.
Volumetric activity concentration	$C_A = A/V$; MBq/mL	Volume of a solution or tissue region.
Weight-normalized administered activity	$k = A_{\text{admin}}/m_{\text{patient}}$; MBq/kg	Patient body mass, not mass of radioactive compound.
Molar activity (supplement)	A/n ; GBq/ μmol	Amount of the specified chemical species.

For a pure radionuclide with molar mass M , $N = mN_A/M$ and

$$a_{s,\text{pure}} = \frac{\lambda N_A}{M}.$$

The units of M must match the desired mass denominator. Added stable carrier decreases activity per mass of the combined material. A high specific or molar activity may reduce the chemical mass administered for a given activity; it does not by itself specify patient absorbed dose.

6.3 The FDG example: retain the arithmetic, correct the rule

The lecture discusses 4 MBq/kg and possible lower activities as a study topic. **Do not interpret 4 MBq/kg as a universal minimum for all FDG examinations.** The EANM tumour-imaging guideline version 3.0 relates administered activity to the imaging system, acquisition duration and patient characteristics. Optimization must preserve diagnostic performance; ethnicity alone is not an adequate activity-selection rule. Adult and paediatric protocols also differ. See the EANM reference in Section 14.

Supplementary explanation / worked extension.

Hypothetical arithmetic only. If an exercise specifies $k = 4$ MBq/kg and a 60 kg patient, then $A_{\text{target}} = 240$ MBq. If a solution is 120 MBq/mL *at the administration reference time*, the ideal volume is

$$V = \frac{A_{\text{target}}}{C_A} = \frac{240}{120} = 2.00 \text{ mL}.$$

If that concentration instead refers to 60 minutes earlier, using a rounded F-18 half-life of 110 min,

$$C_A(60) = 120 \cdot 2^{-60/110} \simeq 82.2 \text{ MBq/mL}, \quad V \simeq 2.92 \text{ mL}.$$

Actual dispensing additionally uses measured activity, timing, residual activity, validated geometry and the authorized protocol. Pre- and post-administration measurements must be decay-corrected to the same reference time before subtraction.

6.4 Activity is not absorbed dose

Activity is measured in Bq; absorbed dose is energy deposited per mass, in gray ($1 \text{ Gy} = 1 \text{ J/kg}$). Equal activities of different radionuclides need not give equal organ doses. Emission energies and yields, spatial distribution, residence time and tissue mass all matter. A dose calibrator principally measures *activity*; it does not directly measure a patient's absorbed dose.

7 Radioactive decay processes

Recording basis: L4, approximately 13:00–64:00. Internal conversion and Auger detail were deferred; their explanations below are supplementary.

7.1 Conservation and a compact comparison

Check charge, nucleon number, total energy and momentum in each complete equation; angular momentum must also be conserved. Ordinary beta and capture equations include a neutrino or antineutrino to complete energy, momentum and lepton accounting. Omitting them may abbreviate a chart label, but it does not give the full physical reaction.

Mode	Daughter A	Daughter Z	Daughter N_n	Primary output
Alpha	$A - 4$	$Z - 2$	$N_n - 2$	Helium nucleus
Beta minus	A	$Z + 1$	$N_n - 1$	$e^- + \bar{\nu}_e$
Beta plus	A	$Z - 1$	$N_n + 1$	$e^+ + \nu_e$
Electron capture	A	$Z - 1$	$N_n + 1$	ν_e ; electron consumed
Gamma	A	Z	N_n	Photon
Internal conversion	A	Z	N_n	Orbital electron ejected
Neutron emission	$A - 1$	Z	$N_n - 1$	Neutron
Proton emission	$A - 1$	$Z - 1$	N_n	Proton

Neutron and proton emission were named but not developed in detail in L4. Nuclear de-excitation can follow an earlier alpha or beta decay, so the categories are not mutually exclusive descriptions of an entire decay scheme.

7.2 Alpha decay



An alpha particle is a helium-4 *nucleus*, containing two protons and two neutrons with charge $+2e$, not initially a neutral helium atom. The daughter is a different element. Alpha emission is common in heavy radioactive nuclei and involves tunneling through the Coulomb barrier.

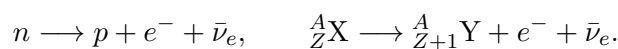
Supplementary explanation / worked extension.

For a specified daughter level, alpha energy is discrete and the daughter recoils. In the nonrelativistic two-body approximation,

$$Q = T_\alpha + T_D, \quad T_\alpha \simeq Q \frac{M_D}{M_D + m_\alpha}.$$

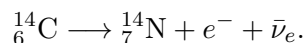
Several daughter levels produce several alpha lines. Excited daughters can subsequently de-excite. Alpha particles deposit energy densely over short ranges, making localization crucial in targeted therapy. Low penetration through intact skin does not imply low hazard after internal uptake.

7.3 Beta-minus decay



The electron is created in the weak transformation; it is not a pre-existing electron stored in the nucleus. Mass number is unchanged because one neutron becomes one proton. The electron's nuclear-equation labels are ${}^0_{-1}\text{e}$: zero nucleons, charge $-e$, but *nonzero rest mass*.

The lecture's example is



Supplementary explanation / worked extension.

Electron kinetic energy is continuous because energy is shared with the antineutrino and recoil. A branch has an endpoint energy; this is not its mean beta energy. Using neutral atomic masses, approximately

$$Q_{\beta^-} = [M(X) - M(Y)]c^2;$$

subtract daughter excitation energy for an excited-state branch. Nuclear medicine beta emitters include I-131, Y-90 and Lu-177; photon emission depends on the full daughter scheme, not the beta-minus label alone.

7.4 Beta-plus decay and annihilation

For an energetically allowed transformation inside a nucleus,

$$p \longrightarrow n + e^+ + \nu_e, \quad {}^A_ZX \longrightarrow {}^A_{Z-1}Y + e^+ + \nu_e.$$

The free isolated proton does not spontaneously undergo this reaction: the nuclear mass balance is essential. The lecture uses ${}^{10}_6\text{C} \rightarrow {}^{10}_5\text{B} + e^+ + \nu_e$. A medically important example is ${}^{18}_9\text{F} \rightarrow {}^{18}_8\text{O} + e^+ + \nu_e$.

Supplementary explanation / worked extension.

With neutral atomic masses,

$$Q_{\beta^+} = [M(X) - M(Y) - 2m_e]c^2.$$

The parent–daughter atomic mass-energy difference must exceed $2m_e c^2 \simeq 1.022$ MeV for an allowed ground-state branch. Excitation adds to the requirement. The positron has a continuous initial kinetic-energy spectrum.

After slowing in matter, a positron usually annihilates with an electron:

$$e^+ + e^- \longrightarrow \gamma + \gamma.$$

For two-photon annihilation approximately at rest, each photon has about 511 keV and the directions are approximately opposite. PET detects these photons in coincidence. It does not directly detect a 511 keV photon emitted by the F-18 nucleus. Positron range and non-collinearity limit spatial resolution. Annihilation in flight and three-photon channels mean that “always exactly two photons at exactly 180°” is too absolute.

7.5 Electron capture and atomic relaxation

$$p + e^- \longrightarrow n + \nu_e, \quad {}^A_ZX + e^- \longrightarrow {}^A_{Z-1}Y + \nu_e.$$

Capture generally involves an occupied bound orbital with wavefunction overlap at the nucleus, commonly the K shell; other shells also contribute. It is a *weak* transformation, not simply electrostatic attraction switching off nuclear forces. Like beta plus, it decreases Z by one and increases N_n by one at fixed A .

The lecture gives carbon-11 capture to boron-11. This is a possible minor branch; carbon-11 is primarily a positron emitter. Iodine-123 is a clearer medically relevant capture example:

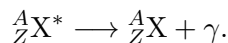
$${}^{123}_{53}\text{I} + e^- \longrightarrow {}^{123}_{52}\text{Te}^* + \nu_e.$$

The daughter can subsequently emit nuclear photons or conversion electrons. Capture also leaves an atomic vacancy. Filling a vacancy releases electronic transition energy through characteristic X-rays or nonradiative electron emission, potentially followed by a cascade.

Supplementary explanation / worked extension.

The total atomic mass release is approximately $[M(X) - M(Y)]c^2$. Energy available to the neutrino in a specified channel is reduced by nuclear excitation, atomic vacancy energy and recoil. Capture has no positron-creation threshold and can occur when beta plus is forbidden, if the chosen final state is accessible. Neutrino energy is approximately discrete for a specified daughter level and captured shell.

7.6 Gamma emission and metastable nuclei



The nucleus loses excitation without changing proton or neutron number. Photon energy is approximately the level-energy difference, with a small recoil correction. A ground-state daughter need not be stable against later decay.

The important example is ${}^{99m}\text{Tc}$ de-excitation to ${}^{99}\text{Tc}$, with a prominent photon near 140.5 keV. The daughter is *Tc-99*, not *Tc-99m*, and remains long-lived and radioactive. Likewise, the familiar 662 keV line associated with Cs-137 arises from de-excitation of Ba-137m populated after Cs-137 beta decay.

Precision note. Gamma rays are distinguished from characteristic X-rays by their nuclear rather than electronic origin, not by a universal energy cutoff. Their energy ranges overlap. Annihilation radiation is another distinct origin of photons.

7.7 Internal conversion and Auger electrons

Supplementary explanation / worked extension.

Internal conversion transfers nuclear transition energy directly to an orbital electron, which is ejected:

$$T_{\text{IC}} \simeq E_{\text{transition}} - B_{\text{shell}} - E_{\text{recoil}}.$$

The nucleus keeps the same A and Z . There is no real emitted gamma photon that must first be reabsorbed. Conversion competes with gamma emission and produces electron lines determined by the transition and shell. The vacancy can subsequently generate X-rays or Auger electrons. The conversion coefficient is $\alpha_{\text{IC}} = N_{\text{conversion}}/N_{\gamma}$ for the specified transition.

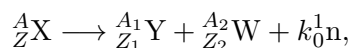
An Auger process is *atomic* relaxation: energy from filling one vacancy ejects another electron. It is a nonradiative competing relaxation route, not necessarily an X-ray emitted and then trapped in the atom. In a simple shell-energy description,

$$T_{\text{Auger}} \simeq B_i - B_j - B_k,$$

with more accurate calculations accounting for changed electron configurations. Conversion electrons originate from nuclear transition energy; Auger electrons originate from atomic rearrangement; beta electrons originate in a weak nuclear transformation.

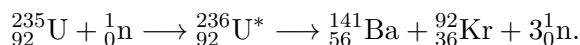
7.8 Spontaneous and neutron-induced fission

Spontaneous fission requires no incoming projectile:



with $A = A_1 + A_2 + k$ and $Z = Z_1 + Z_2$ for this simplified channel. Fragments, neutron multiplicity and emitted radiation vary. Cf-252 has a spontaneous-fission branch, although its dominant decay is alpha emission.

The lecture's uranium example is instead **neutron-induced fission**:



This is one illustrative product channel, not the unique outcome. Charge is $92 = 56 + 36$ and nucleon number is $236 = 141 + 92 + 3$. The mass-92, $Z = 36$ fragment is *krypton*, not plutonium. Additional energy appears in fragment kinetic energy and radiation.

A moderator primarily slows neutrons; control rods primarily absorb neutrons to control reactivity. The emitted fission neutrons are initially fast, not automatically thermal. A chain reaction is not the same concept as a sequential radioactive decay chain.

7.9 Weak reactions are not interchangeable reverse arrows

Precision note. Ordinary beta-minus decay does not require an incoming neutrino. The process $\nu_e + n \rightarrow p + e^-$ is neutrino capture, a separate reaction. Positron capture and reverse weak reactions have their own incoming particles and energy requirements. Do not reverse a partial equation with missing neutrinos and call it the same spontaneous decay.

8 Supplement: clearance, integrated activity and generators

Supplementary explanation / worked extension.

These ideas extend the foundations. Biological/effective half-lives and production were previewed in L1; daughter growth was mentioned in L3. They were not fully derived in the four recordings.

8.1 Physical, biological and effective half-life

Physical half-life describes nuclear transformation. Biological half-life describes removal from a defined compartment. With independent single-exponential clearance and decay and no continuing uptake,

$$\frac{dN}{dt} = -(\lambda_p + \lambda_b)N, \quad \lambda_{\text{eff}} = \lambda_p + \lambda_b,$$

$$\boxed{\frac{1}{T_{\text{eff}}} = \frac{1}{T_p} + \frac{1}{T_b}}, \quad \boxed{T_{\text{eff}} = \frac{T_p T_b}{T_p + T_b}}.$$

For finite positive T_p, T_b , T_{eff} is shorter than either. If $T_p = 6$ h and $T_b = 12$ h, $T_{\text{eff}} = 4$ h. Removal from an organ does not mean the radionuclide has decayed; it may move elsewhere. Uptake, recirculation and multiple compartments invalidate a one-exponential description.

8.2 Time-integrated activity

For $A(t) = A_0 e^{-\lambda t}$,

$$\tilde{A}(0, t) = \int_0^t A(t') dt' = \frac{A_0}{\lambda} (1 - e^{-\lambda t}), \quad \tilde{A}(0, \infty) = \frac{A_0}{\lambda} = A_0 \tau.$$

With activity in Bq and time in seconds, integrated activity counts expected nuclear transformations. In a clearance compartment use the *effective* decay of compartmental activity, yielding A_0/λ_{eff} . Absorbed-dose calculation additionally requires emitted energy and the fraction deposited in each target, not merely integrated activity.

8.3 Parent–daughter growth

For a parent P feeding a daughter D with fraction b ,

$$\frac{dN_P}{dt} = -\lambda_P N_P, \quad \frac{dN_D}{dt} = b\lambda_P N_P - \lambda_D N_D.$$

If $N_D(0) = 0$ and $\lambda_D \neq \lambda_P$,

$$N_D(t) = \frac{b\lambda_P N_P(0)}{\lambda_D - \lambda_P} (e^{-\lambda_P t} - e^{-\lambda_D t}).$$

Multiplying by λ_D gives daughter activity. It can initially rise while parent activity falls. For a much longer-lived parent, secular equilibrium approaches $A_D \simeq bA_P$ after several daughter half-lives. When the parent is longer-lived but not overwhelmingly so, transient equilibrium may arise; the asymptotic ratio is $b\lambda_D/(\lambda_D - \lambda_P)$.

In a Mo-99/Tc-99m generator, Mo-99 (about 66 h half-life) feeds Tc-99m (about 6 h). Chemical elution separates technetium from molybdenum. Parent remaining in the generator produces more daughter. With ideal complete daughter removal, the daughter activity maximum occurs at

$$t_{\max} = \frac{\ln(\lambda_D/\lambda_P)}{\lambda_D - \lambda_P} \simeq 22.8 \text{ h}$$

using 66 h and 6 h. Elution yield, breakthrough and radiochemical quality require actual quality-control measurements.

8.4 Production routes

Reactors produce radionuclides through neutron activation or fission-product routes; cyclotrons use accelerated charged particles in nuclear reactions; generators exploit parent–daughter separation. A production route must be checked for the specific nuclide. It is not sound to assume I-123, I-124 and I-131 are all inevitably co-produced in one reactor extraction. Radionuclidic purity concerns other radionuclides; radiochemical purity concerns chemical forms; chemical purity concerns nonradioactive contaminants.

9 Supplement: from emissions to an image

Supplementary explanation / worked extension.

The following is a study preview of the imaging topics announced in L1, not a claim that they were taught in full.

9.1 Branching fractions and emission yields

If competing channels have partial decay constants λ_i , then

$$\lambda_{\text{total}} = \sum_i \lambda_i, \quad b_i = \lambda_i/\lambda_{\text{total}}, \quad A_i = b_i A_{\text{total}}.$$

The parent has one overall disappearance half-life, $\ln 2/\lambda_{\text{total}}$. A partial half-life is $T_{1/2}/b_i$. A branch fraction is not automatically the yield of a particular photon line: cascades can emit multiple photons and internal conversion can compete with photon emission.

9.2 Activity, photon yield and count rate

If y is photons in the chosen emission line per transformation and ϵ is the total relevant detection efficiency,

$$R_{\text{net}} \simeq A y \epsilon.$$

For 20 MBq, $y = 0.80$ and $\epsilon = 0.001$, the ideal photon emission rate is 16×10^6 photons/s and detected rate is 16000 cps. Activity, photon rate and count rate differ. Background, attenuation, scatter, dead time and the event-selection rules must be accounted for in a real measurement.

9.3 Interaction and attenuation basics

Photoelectric absorption transfers a photon's energy to a bound electron and leaves a vacancy. Compton scattering transfers part of the energy to an electron and redirects the photon. For a monoenergetic narrow beam through uniform material,

$$I(x) = I_0 e^{-\mu x}.$$

μ is an attenuation coefficient in inverse length. The half-value layer is $\ln 2/\mu$, which has length units; physical radioactive half-life has time units. Scatter entering a broad-beam detector complicates this simple attenuation model. Attenuation correction is essential to reliable quantitative emission imaging.

9.4 Planar imaging, SPECT and PET

A gamma camera uses a collimator to select photon directions and detector signals to estimate interaction positions and energies. Better geometric collimator resolution usually trades against sensitivity. Planar imaging superimposes activity through depth. SPECT acquires projections around the subject and reconstructs a tomographic activity distribution. Filtered backprojection and iterative methods are reconstruction approaches, not simply fixed properties of particular manufacturers.

PET uses coincidence lines of response between detected annihilation photons. True coincidences arise from the same annihilation; scattered coincidences include deflected photons; random coincidences combine unrelated events. PET/CT and SPECT/CT combine functional and anatomical information, with CT commonly supporting attenuation correction. Motion or misregistration can create artifacts.

For PET time of flight,

$$\Delta x = \frac{c\Delta t}{2},$$

where the sign depends on which detector's time is subtracted. Timing narrows probable location along a line; it does not eliminate the need for image reconstruction.

9.5 Statistics and quantification

For ideal independent counting with C expected counts, Poisson standard deviation is approximately $\sqrt{C}$, so relative uncertainty is about $1/\sqrt{C}$. Four times the counts halve this component of relative noise. Background subtraction adds variance from both measurements. Dead time and correction/reconstruction methods can change the simple statistical model.

A body-weight-normalized SUV compares regional activity concentration with net administered activity per body mass, with consistent units and decay reference time:

$$\text{SUV}_{bw} = \frac{C_{\text{tissue}}}{A_{\text{net}}/m_{\text{body}}}.$$

Using Bq/g in numerator and denominator gives a dimensionless ratio. If imaging concentration is in Bq/mL, an approximately unit-density convention is commonly implicit. SUVs depend on uptake interval, glucose state, calibration, reconstruction, partial-volume effects and other conditions. They are not intrinsically diagnosis-specific.

9.6 Diagnostic and therapeutic roles

The same nuclear physics supports different clinical goals. Imaging favors emissions that escape tissue and can be detected with useful spatial and energy information. Therapy seeks energy deposition in the intended target while limiting irradiation elsewhere. A radionuclide can emit both therapeutic particles and useful imaging photons.

Role	Illustrative radionuclides	Main physical connection
Single-photon imaging	Tc-99m, I-123	Detectable photons from nuclear de-excitation.
PET	F-18, C-11	Positron emission followed by annihilation.
Beta-particle therapy	I-131, Y-90, Lu-177	Electron energy deposited over a finite range.
Alpha-particle therapy	Ra-223; Ac-225 in targeted-therapy development	Dense energy deposition over short tracks; daughter recoil and distribution matter.
Auger-based approaches	Suitable electron-capture emitters	Highly local electron cascades; subcellular localization matters.

Theranostics links diagnostic targeting information with a corresponding therapeutic strategy, often using different radionuclides attached to related targeting molecules. It does not require one radionuclide to serve both purposes. Current approval, availability and indications vary; this table is a physical comparison rather than a treatment-selection guide.

9.7 QA/QC and radiation protection

QA is the wider system ensuring reliable work; QC includes measurements that check performance. Examples include activity-meter constancy and geometry response, camera uniformity and energy response, SPECT centre-of-rotation alignment, and PET normalization and calibration checks. Frequencies and acceptance limits must follow the equipment, protocol and applicable guidance rather than a generic list.

Patient medical exposures are governed by justification and optimization; occupational/public dose limits are not patient diagnostic dose caps. Diagnostic reference levels help identify protocols needing review and are not individual patient limits. Time, distance and appropriate shielding reduce exposure; contamination monitoring addresses spread of radioactive material. These principles do not justify lowering activity below what the clinical task requires.

10 Supplement: less common decay variants

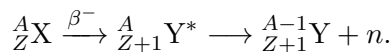
Supplementary explanation / worked extension.

This section retains broader study coverage beyond the recordings without treating rare processes as routine imaging mechanisms.

10.1 Particle emission and delayed emission

Neutron emission gives $(A - 1, Z)$; proton emission gives $(A - 1, Z - 1)$. A neutron-unbound state can emit a neutron without a Coulomb barrier; a proton-emitting state encounters a Coulomb barrier. Neutron-rich nuclei do not all emit neutrons: ordinary beta-minus decay is common.

Beta decay can first populate an excited daughter above a particle threshold, followed by particle emission:



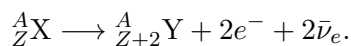
The first step also emits $e^- + \bar{\nu}_e$. The delay reflects the parent weak decay. Two-proton and two-neutron emission may be direct/correlated or sequential, depending on accessible states. Cluster emission ejects a bound cluster heavier than an alpha, leaving $(A - a, Z - z)$ for an emitted ${}_z^a\text{C}$. These require nuclide-specific evaluated data.

10.2 Other de-excitation channels

An energetic nuclear transition can create an electron–positron pair internally when sufficient energy is available (above approximately 1.022 MeV, with recoil accounted for). This is distinct from beta plus: A and Z remain unchanged. It is also distinct from an already propagating photon producing a pair in matter. Rare two-photon nuclear de-excitation is another electromagnetic route.

10.3 Double-weak and electron-occupancy effects

Two-neutrino double-beta-minus decay has the form



It is a second-order weak process rather than two ordinary decays through a populated intermediate state. Double electron capture changes Z by -2 and consumes two electrons. Positron/capture combinations have their own mass thresholds and final-state requirements. These are extremely slow processes, not routine nuclear medicine production or imaging mechanisms.

A proposed neutrinoless double-beta mode would omit the antineutrinos and violate lepton number; it must be kept distinct from established two-neutrino modes. No experimental discovery claim is made here. In bound-state beta-minus decay, the newly produced electron occupies an available bound orbital, particularly relevant to highly ionized atoms. Radiative weak-decay variants include an additional photon. These qualifications explain why the simplest neutral-atom decay model is not universal.

11 Corrections and unresolved transcript details

The corrections below identify important differences between transcript wording and the scientific statements used in this edition. They do not establish whether an error originated with speech, transcription or an unseen slide.

Recording	Wording / issue	Treatment in these notes
L1 75–77 min	Quantum-number orientations; quark fractions called spin	General m_ℓ range given; $2/3$ and $-1/3$ identified as charges.
L2 10–35 min	Weak for unstable, strong for stable; equal force strengths	Interaction-based description; dominance separated from relative strength.
L2 24–25 min	Higgs as gauge carrier / validation of relativity	Higgs field role distinguished from gauge bosons and general relativity.

Recording	Wording / issue	Treatment in these notes
L2 55–58 min	Carbon-12 reference explained by isolated-atom occurrence	Reference definition stated without the incorrect chemical rationale.
L2 about 70 min	“Cesium-157” with about 30-year half-life	Consistent identification is Cs-137; no invented Cs-157 data.
L2 76–84 min	Abrupt force switch; alpha poisoning linked to neutrons	Finite-range interactions explained; alpha emission does not imply neutrons.
L2 about 88 min	Speed of light exponent / unit errors	$c \simeq 3 \times 10^8$ m/s.
L2 92–93 min	Helium-4 target 27.27 MeV	Computed about 28.30 MeV; slide comparison remains unavailable.
L3 12–24 min	“Multiply by e / $\ln$ ”	Exponentiation / logarithm operations shown explicitly.
L3 32–35 min	Activity can never be zero	Continuous expectation separated from finite populations and measured counts.
L3 38–40 min	Daughter terminology attributed to Marie Curie	Unsupported etymology omitted; parent/daughter/progeny retained as terminology.
L4 6–9 min	Mean life as isotope-mixture half-life	$\tau = 1/\lambda$; mixture described as a sum of exponentials.
L4 10–13 min	Specific activity used for patient MBq/kg; 4 MBq/kg minimum	Distinct denominators; guideline- and scanner-dependent activity.
L4 20–38 min	Mass number and atomic number interchanged; electrons described as massless	A counts nucleons, Z protons; electron mass nonzero.
L4 35–47 min	Inner electrons called valence; EC as electrostatic; Auger via trapped X-ray	Inner/valence distinction, weak capture, and nonradiative atomic relaxation.
L4 48–56 min	Stable gamma daughter; Tc-99m daughter label; spontaneous fission example	Lower-energy may remain radioactive; Tc-99 daughter; induced fission distinguished.
L4 55–57 min	Plutonium-92 at $Z = 36$; emitted thermal neutrons; moderator regulates by analogy	Kr-92; initial fast neutrons; moderator slowing distinguished from control-rod absorption.
L4 59–60 min	Odd–odd list ends “nitrogen-4”	Nitrogen-14; hydrogen example identified as deuterium.

Course administration is time-sensitive. The orientation settled on 45% examinations and 5% attendance, while the initially spoken group-component figures are internally inconsistent if all treated as separate components. The current syllabus should resolve the group weighting; these notes do not invent a 100% grading scheme. L4 announced online meetings on 21 and 24 September and an intended in-person return on 28 September, subject to announcements. These are historical recording statements, not verified current scheduling instructions.

12 Practice with worked answers

Supplementary explanation / worked extension.

1. **Identify the nucleus.** A nuclide has $A = 131$, $Z = 53$. How many neutrons?
 $N_n = 131 - 53 = 78$. It is iodine-131. A neutral atom has 53 electrons; ionization can change electron number without changing these nuclear labels.
2. **Specific activity from a pure isotope.** Find the ideal activity per gram for a hypothetical pure radionuclide with 6 h half-life and molar mass 99 g/mol.
 $\lambda = 0.693147/21600 = 3.2090 \times 10^{-5} \text{ s}^{-1}$. Then $a_s = \lambda N_A/M \simeq 1.952 \times 10^{17} \text{ Bq/g}$. A real carrier-containing solution can have a much smaller value.
3. **Unit conversion.** Convert 5 mCi to MBq and dpm.
 $5 \times 37 = 185 \text{ MBq}$. Multiplying $185 \times 10^6 \text{ Bq}$ by 60 gives $1.11 \times 10^{10} \text{ dpm}$. This is transformations per minute, not detected counts per minute.
4. **Partial half-life interval.** A 6 h source starts at 600 MBq. What remains after 9 h?
 $A = 600 2^{-9/6} = 212.13 \text{ MBq}$. Non-integer numbers of half-lives are valid.
5. **Mean life.** A source has $\lambda = 0.20 \text{ h}^{-1}$. Find mean life and half-life.
 $\tau = 5.00 \text{ h}$ and $T_{1/2} = 0.693147/0.20 = 3.466 \text{ h}$.
6. **Mixture.** Two independent components each start at 100 MBq and have half-lives 2 h and 6 h. Find total activity after 6 h.
 $A = 100/2^3 + 100/2 = 12.5 + 50 = 62.5 \text{ MBq}$. Using an arithmetic mean half-life of 4 h would give a different, incorrect result.
7. **Integrated activity.** A single decay-only source starts at 100 MBq with a 6 h half-life. Find transformations during the first 6 h.
 $\tilde{A} = (100 \times 10^6)/(3.2090 \times 10^{-5}) \times (1 - 1/2) \simeq 1.558 \times 10^{12} \text{ transformations}$. Convert hours to seconds before combining with Bq.
8. **Net administered activity.** Pre-administration syringe activity is 250 MBq at 10:00. Residual is 10 MBq at 10:10. Using a 110 min half-life, calculate net activity referred to 10:00.
Residual back-corrected to 10:00 is $10 2^{10/110} = 10.650 \text{ MBq}$; net is 239.350 MBq. Both values need the same reference time.
9. **Effective half-life.** $T_p = 8 \text{ h}$ and $T_b = 24 \text{ h}$.
 $T_{\text{eff}} = 8 \times 24/(8 + 24) = 6 \text{ h}$. In the stated model compartmental activity falls to one quarter after 12 h.
10. **Alpha daughter.** $^{210}_{84}\text{Po}$ emits an alpha.
The daughter has $A = 206$, $Z = 82$: $^{206}_{82}\text{Pb}$. Include ^4_2He on the product side.
11. **Beta and capture.** Compare ^{14}C beta minus with ^{18}F beta plus.
C-14 gives N-14, an electron and an antineutrino. F-18 gives O-18, a positron and a neutrino. Neither process changes mass number.

12. **Threshold test.** A hypothetical neutral-atom mass difference is $0.80 \text{ MeV}/c^2$. Is ground-state beta plus possible?
No: $Q_{\beta+} = 0.80 - 1.022 = -0.222 \text{ MeV}$. Electron capture may be possible if its nuclear and atomic final state is accessible; it must be checked separately.
13. **Conversion electron.** A 200 keV nuclear transition converts an electron bound by 30 keV. Its kinetic energy is approximately 170 keV, neglecting recoil. This is not a continuous beta spectrum.
14. **Count precision.** Compare 10000 counts with 40000 counts.
Ideal relative Poisson uncertainties are 1% and 0.5%. Doubling counts alone improves relative uncertainty by $\sqrt{2}$, not a factor of two.
15. **Chart direction.** On horizontal N_n , vertical Z , what decay moves one square left and one up?
Beta minus: $(\Delta N_n, \Delta Z) = (-1, +1)$. Recheck axes before using directional mnemonics.
16. **Why is 511 keV not the beta-plus endpoint?**
It is approximately the energy of each photon from at-rest two-photon annihilation. Initial positron kinetic energies are distributed continuously according to the beta-decay branch.
17. **Does a ground-state daughter mean no further activity?**
No. Tc-99 is the lower nuclear state after Tc-99m de-excitation, yet is itself radioactive. “Ground state” and “stable nuclide” describe different properties.
18. **Can all nuclei in a small sample have decayed?**
Yes. The nonzero exponential curve is the expected population, not a guarantee that at least one parent survives in every finite realization.

13 One-page equation and distinction sheet

$$\begin{array}{ll}
 N_n = A - Z, & R \simeq r_0 A^{1/3}, \\
 B = \Delta m c^2, & 1 \text{ u } c^2 \simeq 931.494 \text{ MeV}, \\
 A(t) = \lambda N(t), & \lambda = \ln 2 / T_{1/2}, \\
 A(t) = A_0 e^{-\lambda \Delta t} = A_0 2^{-\Delta t / T_{1/2}}, & \Delta t = \ln(A_0 / A) / \lambda, \\
 \tau = 1 / \lambda = T_{1/2} / \ln 2, & p(t) = \lambda e^{-\lambda t}, \\
 C_A = A / V, & a_s = A / m, \\
 A_{\text{target}} = k m_{\text{patient}}, & V = A_{\text{target}} / C_A, \\
 T_{\text{eff}} = T_p T_b / (T_p + T_b), & \tilde{A}_{\infty} = A_0 / \lambda_{\text{eff}}, \\
 R_{\text{net}} \simeq A y \epsilon, & \sigma_C / C \simeq 1 / \sqrt{C}.
 \end{array}$$

The effective-half-life and integrated-activity expressions assume a single compartment without continued input. Counting expressions are idealizations.

- **Activity versus dose:** Bq counts transformations per second; Gy measures deposited joules per kilogram.
- **Mass number versus atomic number:** A counts nucleons; Z identifies the element.
- **Half-life versus mean life:** 50% surviving at $T_{1/2}$; about 36.8% at τ .
- **Primary versus secondary radiation:** beta plus produces a positron; annihilation produces the PET photons. EC or conversion can be followed by X-rays/Auger electrons.
- **Reference times:** decay-correct activity and residual measurements before comparing or subtracting them.
- **Precision:** check units, use isotope masses, retain guard digits, state approximation and round only the result.

$$1 \text{ Ci} = 37 \text{ GBq}, \quad 1 \text{ mCi} = 37 \text{ MBq}, \quad 1 \text{ Bq} = 60 \text{ dpm}.$$

14 References and verification sources

The lectures provide the topic sequence and questions. The following sources support corrections and further checking. Numerical practice and diagrams are original teaching material, not copied textbook figures. Accessed 18 September 2026. Exact decay schemes should be checked in evaluated data rather than inferred from a simplified example.

1. IAEA, *Nuclear Medicine Physics: A Handbook for Teachers and Students* (2014), particularly Chapter 1: atomic/nuclear physics, binding energy, decay and decay chains. [Full handbook](#).
2. IAEA Nuclear Data Section, *LiveChart of Nuclides*. [Nuclide-chart entry page](#). Use for nuclide-specific half-lives and branches; chart axes must be inspected.
3. NNDC, Brookhaven National Laboratory, *NuDat 3: Beta Decay*. [Weak transformations, Q-values and classification](#). [NuDat evaluated-data interface](#).
4. NIST, *Atomic Weights and Isotopic Compositions for All Elements*. [Isotopic atomic-mass table](#). Used for H-1, He-4, Li-7, O-16 and U-235 exercise inputs. [Fundamental constants](#).
5. CERN, *The Standard Model*. [Particles, interactions and limits of the model](#). CERN, *What's so special about the Higgs boson?* [Higgs mechanism and discovery](#).
6. Boellaard et al., [^{18}F]/FDG PET/CT: EANM procedure guidelines for tumour imaging: version 3.0, *The EANM Journal* (2025), 100006. [Guideline and supplementary material](#); [EANM oncology guideline directory](#). Used to qualify the universal-activity claim; activity selection depends on scanner, acquisition and patient factors.
7. NNDC, *Tc-99 isomeric-transition decay, 6.0072 h*, evaluated July 2017. [Evaluated levels and gamma-ray data](#). The 6 h and 140.5 keV figures in these notes are deliberately rounded.
8. IAEA, *Radiation Protection and Safety of Radiation Sources: International Basic Safety Standards*, GSR Part 3. [Medical exposure, optimization and reference levels](#).
9. The course names Cherry, Sorenson and Phelps, *Physics of Nuclear Medicine*, as the main textbook. Consult the edition assigned in the syllabus. This note does not claim that a specific edition or its unseen figures were reviewed.

A Recording identifiers and scope notes

These identifiers allow the private source recordings to be found again; the recordings themselves and participant names are not reproduced.

- L1: of_c43c23ec493b984d9d936abce8e31e3b.
- L2: of_3d9aa268c8c0bd17ece8914d0471a0ee.
- L3: of_92c075a3191c66b0f5a9adb07f8c389b.
- L4: of_2b17d4008cb10aa998c96fd1230c6011.

Transcript coverage was complete for the four selected recordings. Selection was based on the full library's recording titles and the scientific content of the selected transcripts. Unrelated-course recordings were not treated as nuclear medicine sources. Audio, lecture-slide images, chat answers and Canvas materials were not independently available for visual or auditory verification in this synthesis.

Administrative digressions, attendance roll calls, personal anecdotes and speculative analogies were omitted unless they affected scientific interpretation. All important topic-level content from the four recordings is represented. Supplementary imaging, production and uncommon-decay sections are introductory; they do not imply that later course lectures have already occurred.